

FIT CEP ASSIGNMENT

CT-174

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SE-24009

SECTION-A

SOFTWARE ENGINEERING

QUESTION # 1

Explain the Blu-Ray Disc along with its structure.

BLU - RAY DISC

DEFINITION

Blu-ray Disc (BD) is a name of a new optical disc format jointly developed by the Blu-ray Disc Association (BDA). The format was developed to enable recording, rewriting and playback of high-definition video (HD), as well as storing large amount of data. BD is a next generation optical disc format. The name was derived from technology which utilizes a blue-violet laser to read and write data on Blu-ray disc.

ADVANTAGES

- More storage capacity on a disc of same size
- High data transfer rate

- strong content protection
- Higher disc life
- Compatible with analog and digital transmission

STRUCTURE

The structure of Blu-ray disc can be described as;

- Upper Layer of polish - used for any labels on the disc.
- Polycarbonate Layer - 1.1 mm thick disc, on which particular layers are put.
- Recordable Layer - on this layer user data is recorded.
- Translucent Layer - it separates two recordable layers.
- Protective Layer - much thinner layer that protects the layer on which data is protected.
- Hardening Protective Layer - a special layer that protects the carrier against damages.

APPLICATIONS

Main applications of BD are;

- High definition television recording.
- High definition video distribution
- Mass data storage
- Professional Storage

COMPARISON

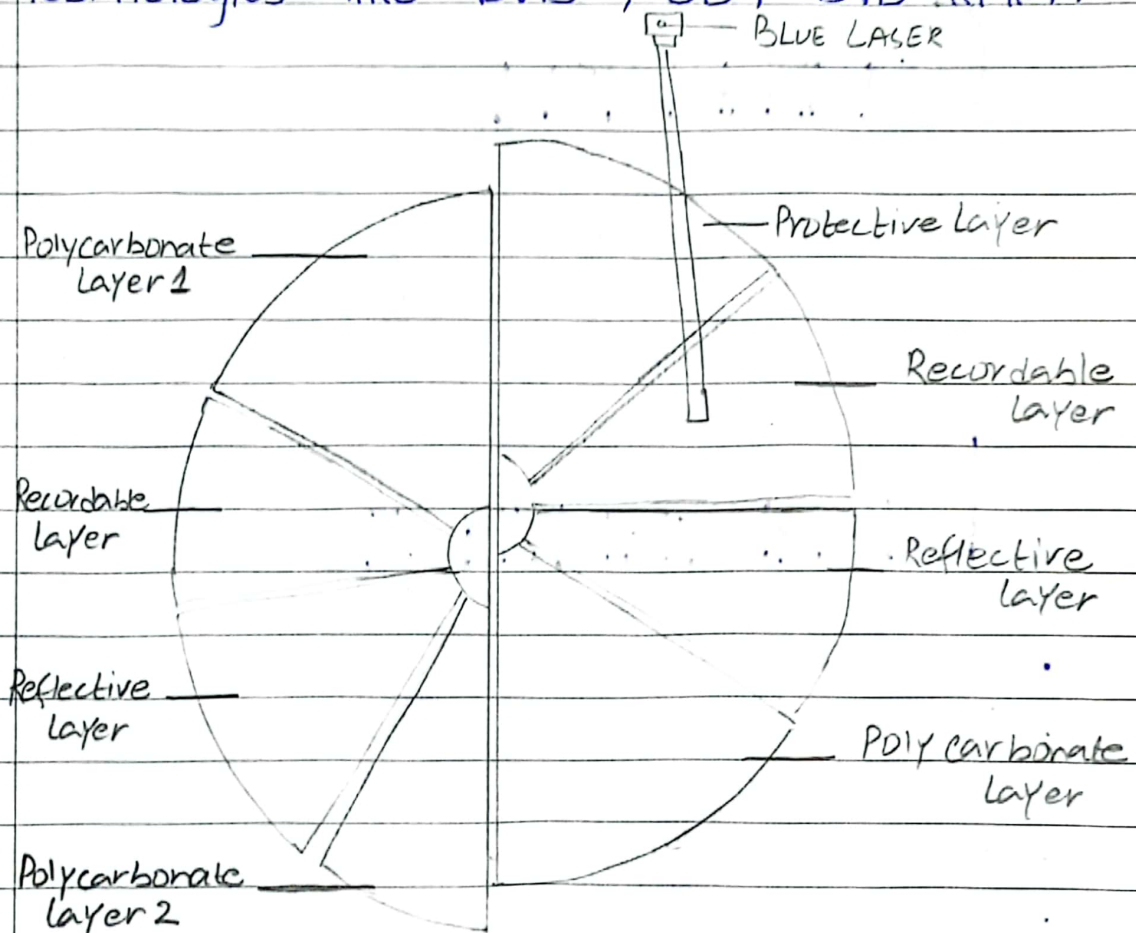
BD offers significantly higher storage capacity compared to CD's and DVD's, making them suitable for storing large files and HD content.

DISADVANTAGES

- very expensive technology
- low compatibility; BD can be played only with Blu-ray players.
- can be easily damaged or scratched.
- loading a film can take minutes as so much data needs to be read.

CONCLUSION

BD has been a consistent road map to emerging disc technologies. It can store upto 54 GB, enough to hold 4.5 hours of HD video. It is very likely to that the technology will be adopted as the next generation optical disc format for PC data storage and replace technologies like DVD, CD, DVD-RAM.



QUESTION #2

Explain various types of storage devices. Explain the structure of magnetic disc.

STORAGE DEVICES

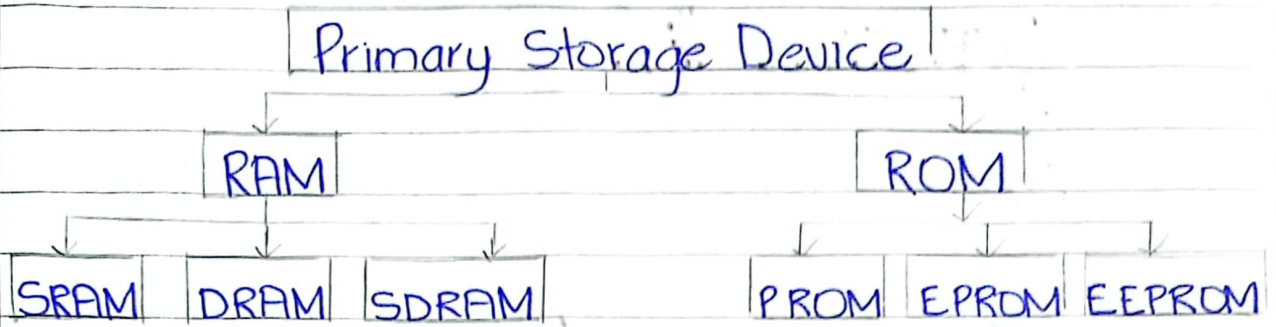
A storage device is an integral part of the computer hardware which stores data to process the result. Without a storage device, a computer would not be able to run or even boot up.

TYPES OF STORAGE DEVICES

Some commonly used storage devices are;

1) PRIMARY STORAGE DEVICE

A primary storage device is a type of data storage that is actively used by the computer's C.P.U. The data in it is volatile (lost when powered off). CPU directly access primary storage.



2) MAGNETIC STORAGE DEVICE

One of the most widely used types of digital data storage is magnetic storage. This refers to magnetized media such as hard disk drives (HDD), that are used by computer system to store data permanently. Commonly used magnetic storage device include magnetic tape, floppy disk etc. They are the most affordable way to store large amount of data.

3) FLASH MEMORY DEVICE

It is a cheaper and more portable storage device. It is the most commonly used device to store data because is more reliable and efficient as compared to other storage devices. Some commonly used flash memory devices are Pen drive, SSD, SD Card etc.

4) OPTICAL STORAGE DEVICE

It is a removable storage device. In it data is written and read with a laser. Typically, data is to optical media such as CD's and DVD's.

5) CLOUD AND VIRTUAL STORAGE

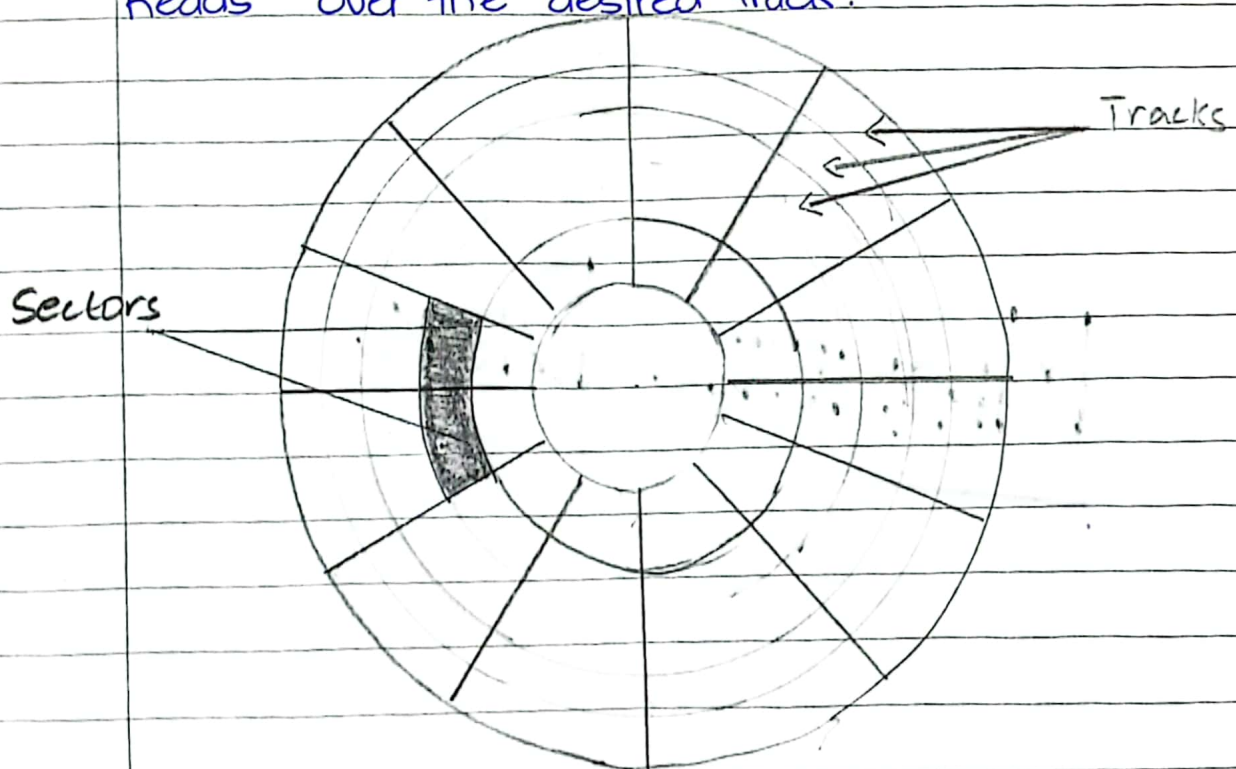
Cloud storage device uses remote servers to save data such as files, data, videos. Users upload data to servers via an internet connection, where it is saved on a virtual machine on a physical server.

MAGNETIC DISK

A magnetic disk is a storage device that uses a magnetization process to write, rewrite and access data. It is covered with a magnetic coating and stores data in form of tracks, spots and sectors.

STRUCTURE

Magnetic disks are flat circular plates of metal or plastic, coated on both sides with iron oxide. Input signals, which may be audio, video, or data are recorded on the surface of a disk as magnetic patterns or spots in spiral tracks by a recording head while the disk is rotated by a drive unit. The data is read or written by a disk drive which rotates the discs and positions the read/write heads over the desired track.



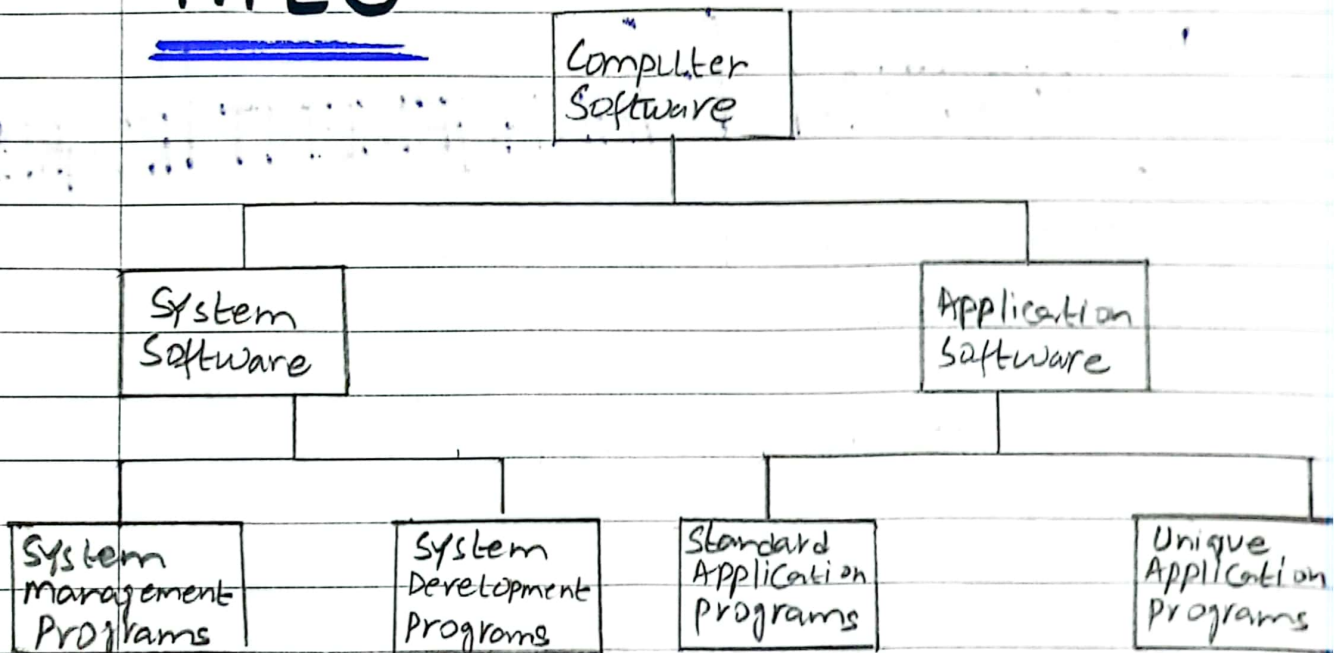
QUESTION # 3

Explain the various types of computer software.

COMPUTER SOFTWARE

Computer software is a collection of instructions, data or computer programs that are used to run machines and carry out activities. It is basically a set of instructions.

TYPES



1. SYSTEM SOFTWARE

System software directly operates the hardware of computer and provide basic functionality to users. It controls the internal functionalities. It is like an interface between hardware and user applications.

TYPES

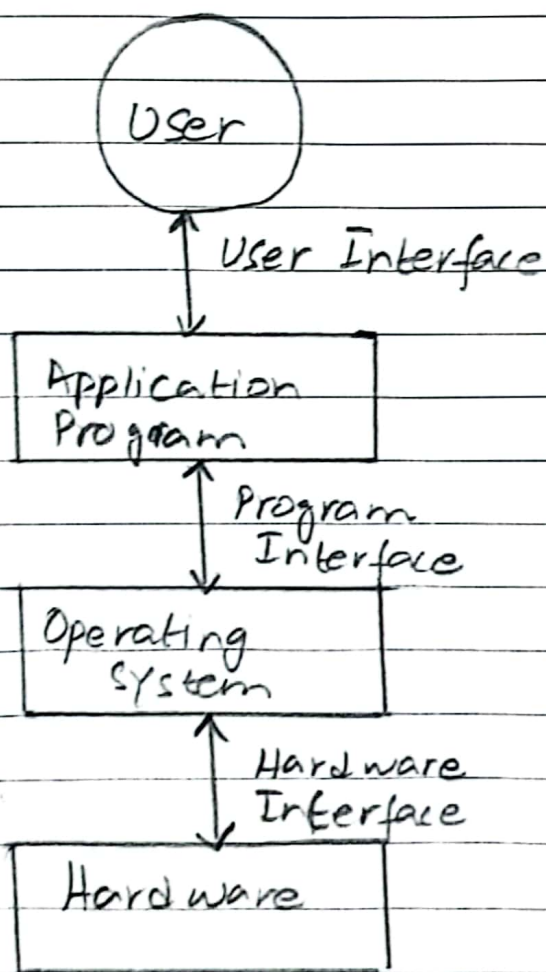
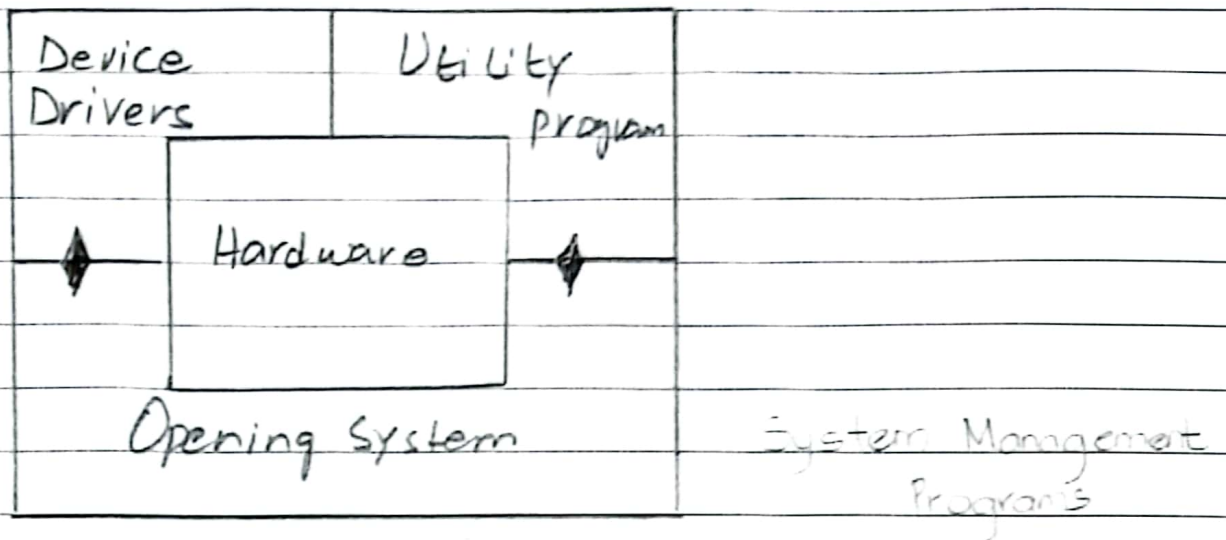
System software is further divided into sub-types.

a. SYSTEM MANAGEMENT PROGRAM

System management programs are those programs that are meant for operating the hardware system and managing their resources effectively. They minimise the human interaction and maximized the productivity of computer system.

System management programs include ;

- Operating System
- Utility Programs
- Device Drivers



OS - User Interface

Operating System - OS is the principal component of system software and is responsible for overall management of computer resources. It provides interface between the computer and the user.

Utility Programs - They enable OS to perform some additional tasks. A utility program is not an essential part of an operating system.

Device Drivers - Device Driver is a special software that enables a hardware device to perform an operation according to command given by end user.

b) SYSTEM DEVELOPMENT PROGRAMS

These programs allow the user to develop programs in different programming languages. They include;

Language Translators - It is a program that converts high-level-language into low-level language program. Compiler, Assembler and Interpreter are the examples.

Debugger - It is a program used to detect errors and bugs in a program.

Linker - A program that combines several object code files into a single executable program.

Editor - Editor is a special program that allow the user to work with text in computer system allowing copy, cut, paste and other operations.

2) APPLICATION SOFTWARE

Application software enables the end user to perform various task, such as creating documents, creating presentations, developing databases etc.

Some various programs are;

Word processor - It is an application software that is generally used to create documents.

Spread Sheets - It is an application software that enable the user to main a worksheet in different cells.

DBMS - It helps to maintain and stores records in a database. With the help of DBMS, end user can store, maintain, update data store in database.

QUESTION # 4

Write a detailed note on UNIX and LINUX OS along with key features of Windows XP.

LINUX OS

Linux is a free and open source operating system. The system's source code is accessible to everyone for anyone to look at and change. It is well-known for being cost effective.

Developed in 1991, Linux OS is a powerful and flexible opensource platform. It acts as the basis of a variety of devices and well known for its reliability, safety and flexibility.

USES

Because it is free, open source and extremely flexible, Linux is widely utilized. For servers and developers, it is the ideal option because it offers strong, security, stability and performance. Due to its adaptability, users can customize the OS.

ADVANTAGES

- It is an open source OS. This means the source code is available for everyone.
- In terms of security, Linux is more secure than any other OS.
- The software updates are easy.
- It maintains the privacy of user.
- The flexibility is high and network friendly.
- It is fast and easy to install.

DISADVANTAGES

- It is not very user friendly.
- It has small peripheral devices (hardware drivers) as compared to windows.

ARCHITECTURE

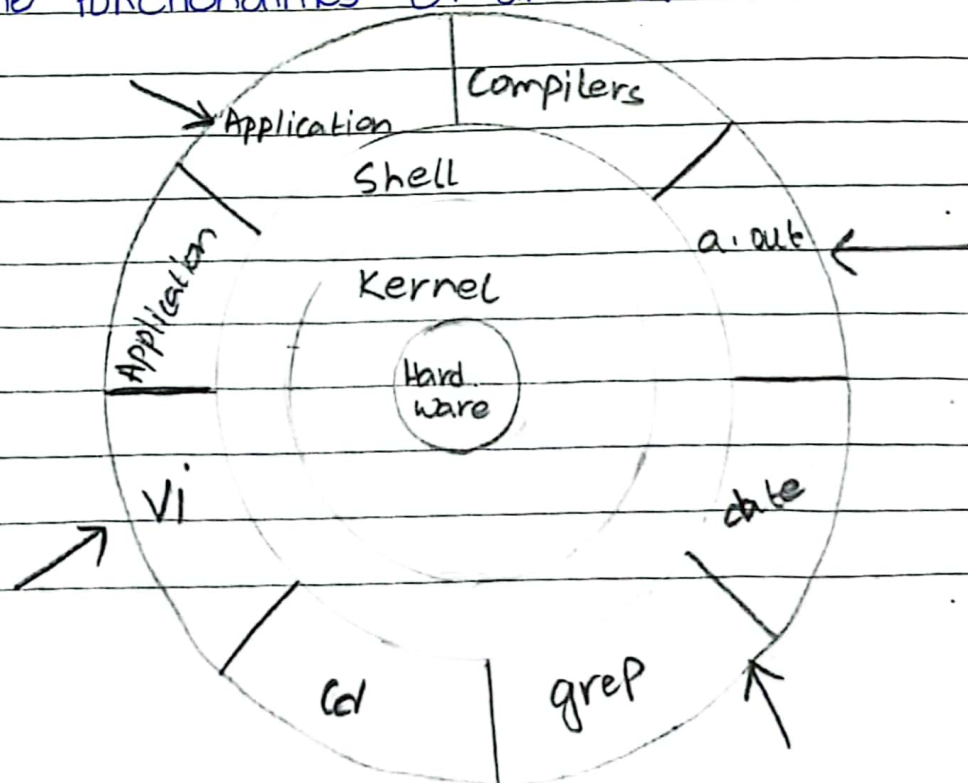
The architecture of Linux OS consist of following layers ;

Hardware Layer - consist of all peripheral devices.

Kernel - It is the core component of OS, interacts directly with hardware.

Shell - An interface to kernel, hiding the complexity of kernel's function from users. It takes commands from users.

Utilities - programs that provide the user most of the functionalities of an OS.



UNIX

Unix is an innovative OS which was developed in 1970's. It is a backbone for many OS. It was designed for developers only, UNIX played a most important role in the development and creation of the software and computing environments. The core part of UNIX lies in its Kernel. It also features a Command Line Interface that allow users to interact with the system through a series of commands.

KEY FEATURES

- **Multiuser Support** - It allows multiple users to simultaneously access the same system and share resources.
- **Multitasking** - It is capable of running multiple processes at the same time.
- **Shell Scripting** - It provides a powerful scripting language to automate task.
- **Security** - It has a robust model for security.

- **Portability** - It can run on wide variety of hardware platforms.
- **Communication** - It supports communication methods using the command.

ADVANTAGES

- **Stability** - It is known for its stability and reliability.
- **Security** - It's security model makes it popular choice for systems that require high levels of security.
- **CLI** - It's CLI's interface allows for powerful and efficient interaction with system.

DISADVANTAGES

- **Complexity** - UNIX can be complex for users who are used to GUI's.
- **Cost** - UNIX systems can be expensive.

- Steep Learning Curve - It requires technical knowledge and expertise.

STRUCTURE

Following is the description of each layer of structure in UNIX;

Layer-1 - Hardware

This layer consist of all hardware related information in the environment.

Layer-2 - Kernel

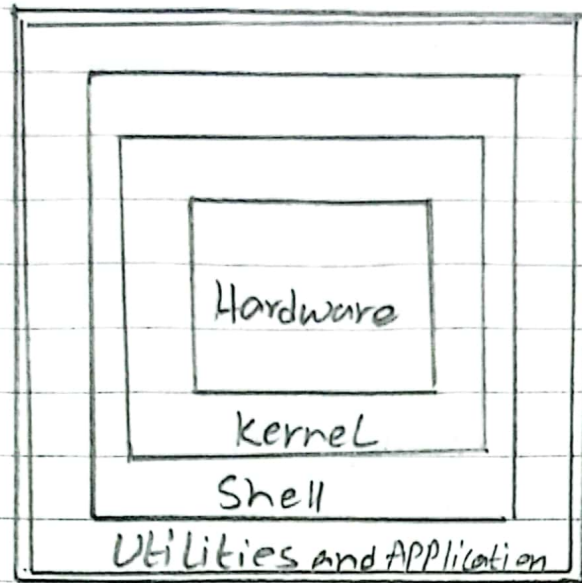
The kernel of UNIX runs on the particular hardware. It is the core of OS. It also works as a device manager.

Layer-3 - Shell

The shell is an interpreter that interpretes command submitted by the user and calls the program.

Layer - 4 - Application Program Layer

It is the outermost layer that executes the given external applications.



WINDOWS XP

It is an OS that allow the user to use different types of applications. It is a graphical user interface. This OS has multi-tasking capabilities, allowing user to view many screens at the same time.

KEY FEATURES

Windows XP was a popular OS released in 2001 by Microsoft. Some key features are;

- User friendly Interface - It was designed to be easy to use.

- **Improved Performance** - It offered better performance than previous versions.
- **Networking** - It included support for WIFI, bluetooth and network file sharing.
- **Remote Desktop Support** - Users could enable remote assistance.
- **Themes** - Users could create custom controls.
- **Compatibility** - Windows XP was compatible with a wide range of hardware and software.
- **Multilingual Support** - Added support for different languages.

Hence, Windows XP allows everyone to have separate computer accounts. It enables you to work with a variety of programs on your computer.